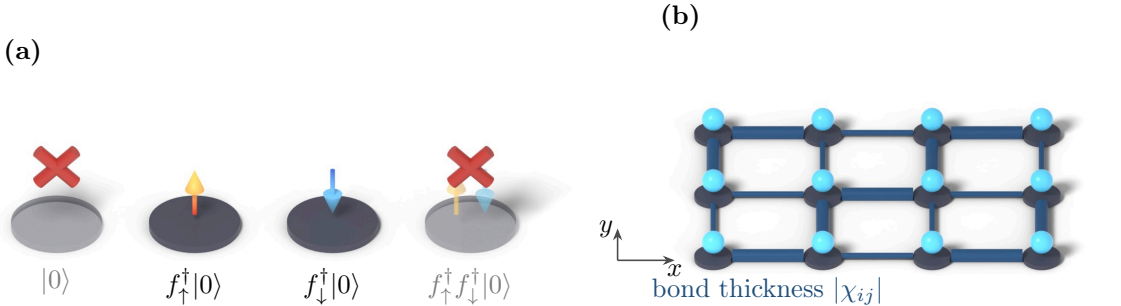

(a)
$|0\rangle$
$f_\uparrow^\dagger|0\rangle$
$f_\downarrow^\dagger|0\rangle$
$f_\uparrow^\dagger f_\downarrow^\dagger|0\rangle$
(b)
y
x
bond thickness $|\chi_{ij}|$